MUTATT:VISUAL-TEXTUAL MUTUAL GUIDANCE FOR REFERRING EXPRESSION COMPREHENSION

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Referring expression comprehension

Input

An image I

A query sentence $\mathcal{E}$:

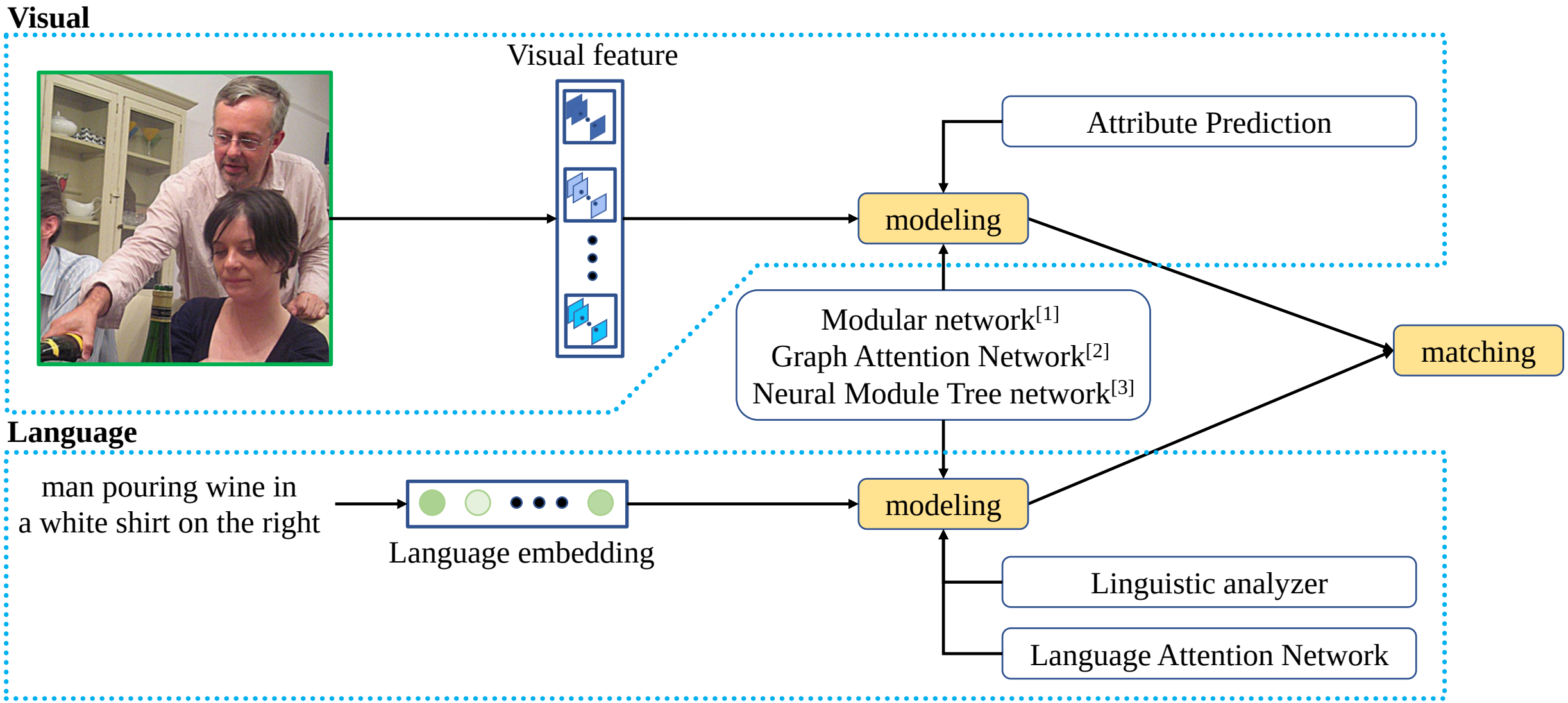
man pouring wine in a white shirt on the right.

retrieval target region
by query sentence

Output

Localize the target region

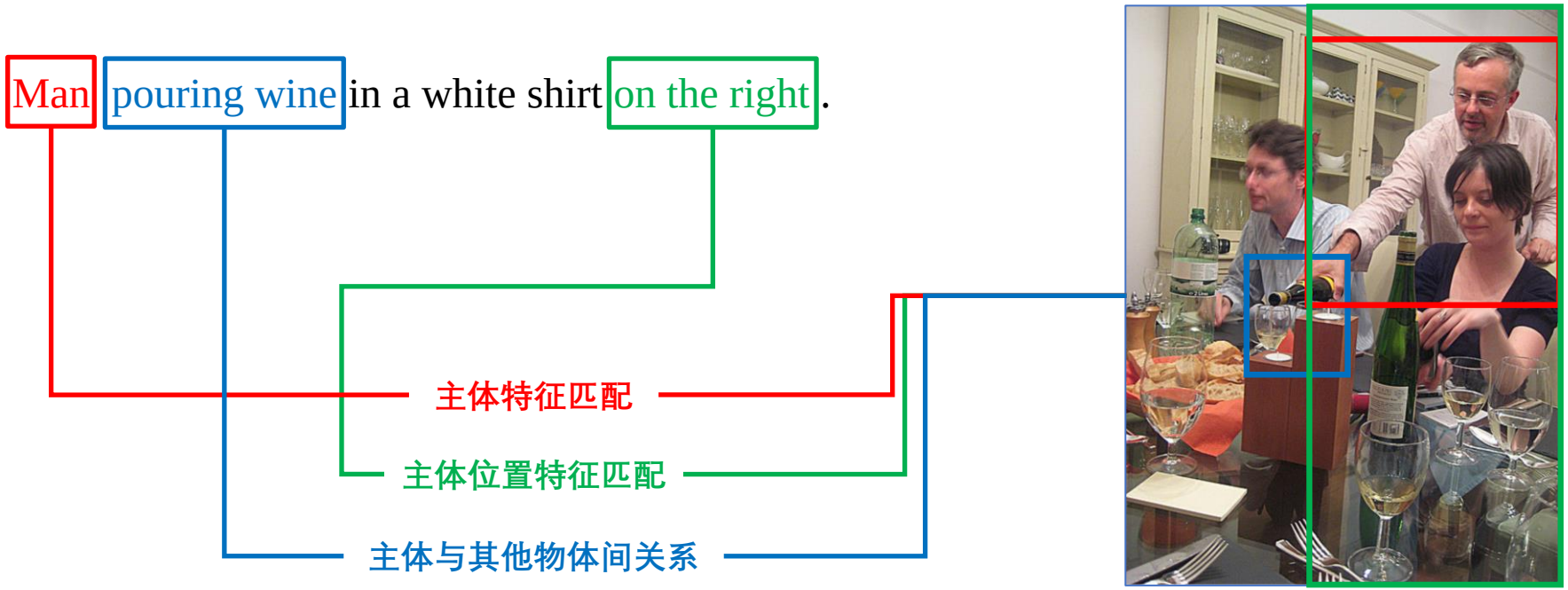
Related work



1. Licheng Yu, Zhe Lin, Xiaohui Shen, Jimei Yang, Xin Lu, Mohit Bansal, and Tamara L Berg, "Mattnet: Modular attention network for referring expression comprehension," in CVPR,2018.
2. Sibei Yang, Guanbin Li, and Yizhou Yu, "Dynamic graph attention for referring expression comprehension," in ICCV, 2019.
3. Daqing Liu, Hanwang Zhang, Feng Wu, and Zheng-Jun Zha, "Learning to assemble neural module tree networks for visual grounding," in ICCV, 2019.

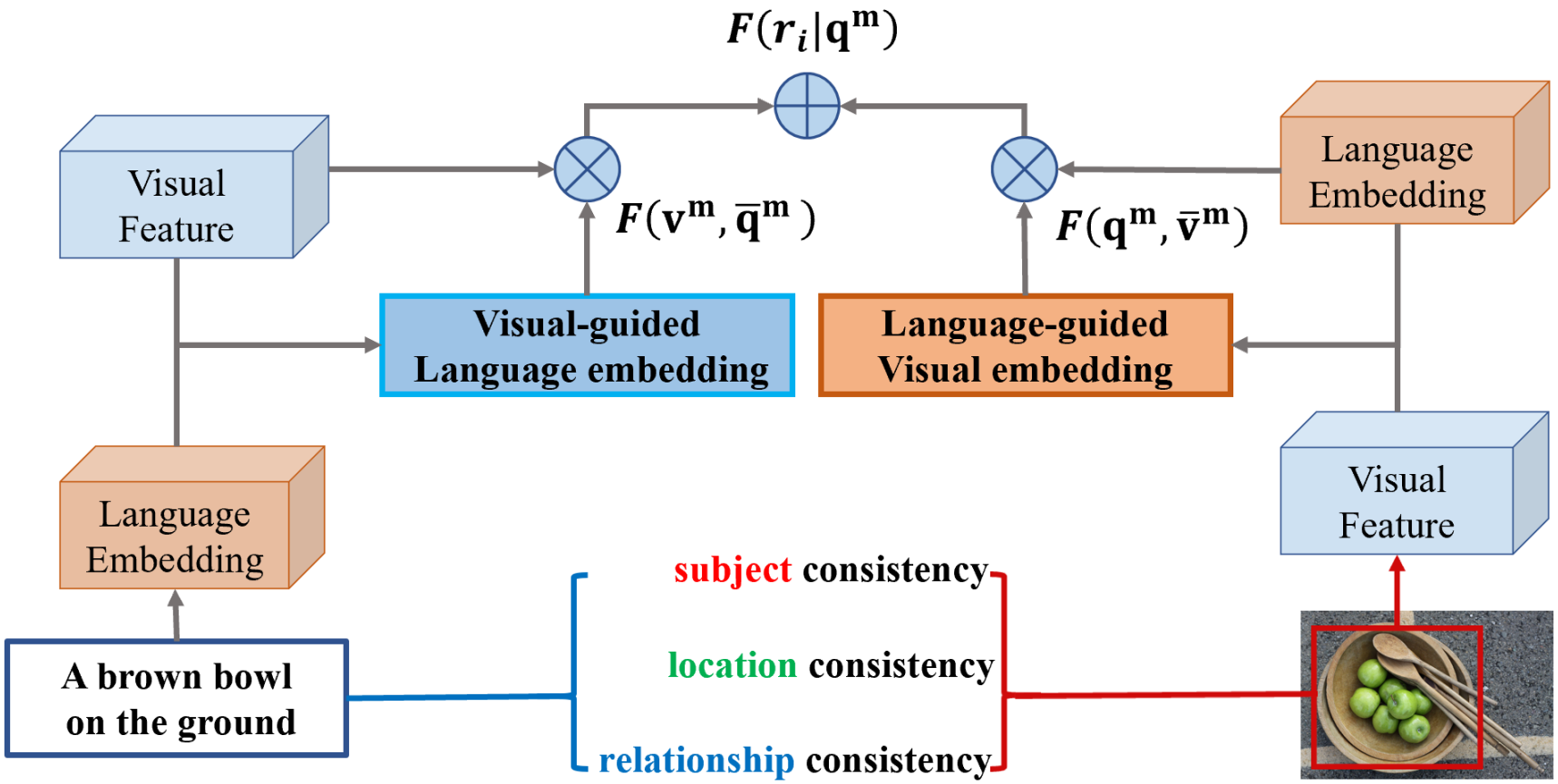
Motivation

- Consistency exists between vision and language.
- Visual and language information play an equally important role.

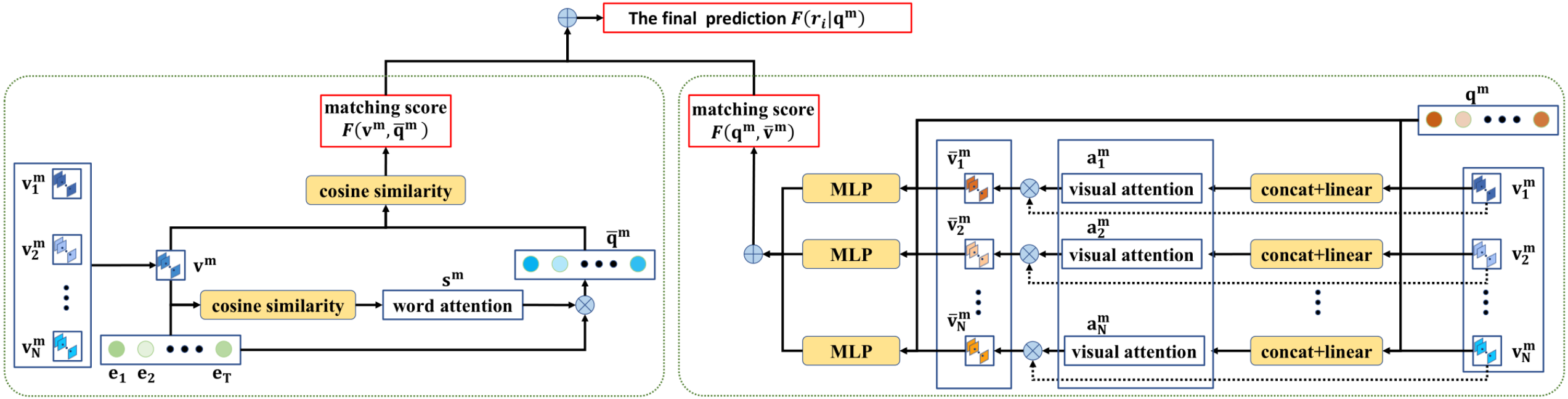


Our method

Vision-textual mutual guidance (MutAtt): visual-guided language embedding and language-guided visual embedding.

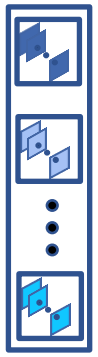


Framework



Visual-guided language embedding

Language-guided visual embedding



visual feature of module m : $\{\mathbf{v}_n^m\}_{n=1}^N$

faster R-CNN + ResNet101

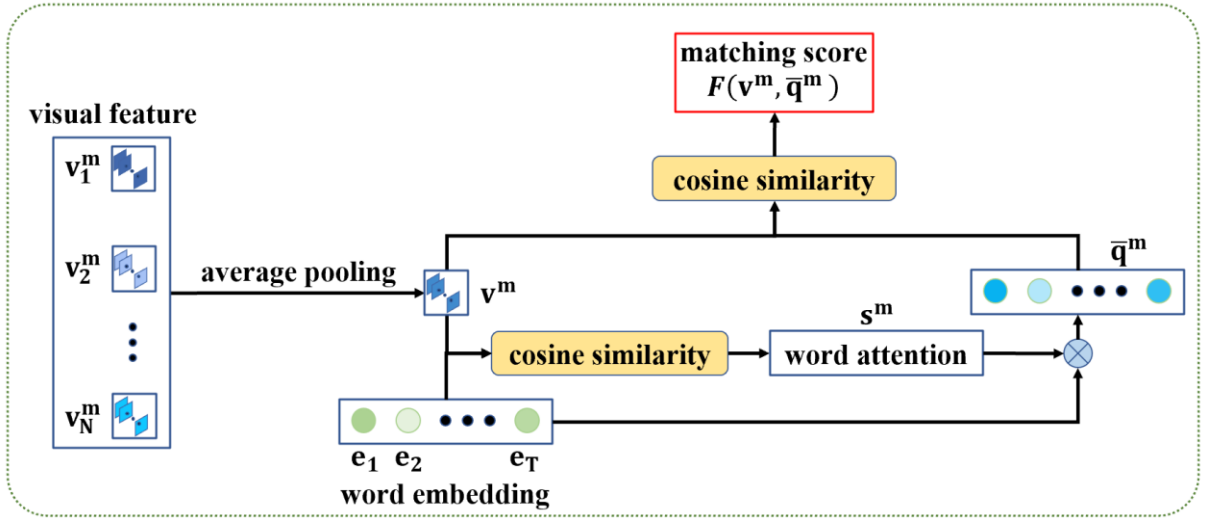
word embedding : $\{\mathbf{e}_t\}_{t=1}^T$

glove embedding

language embedding of module m : $\mathbf{q}^m$

LSTM + language attention network

Visual-guided language embedding



1. Attention from visual feature of module m to the t-th word embedding:

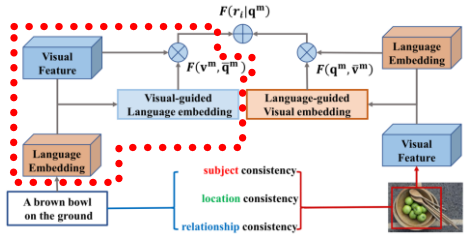
$$\mathbf{s}_t^m = \frac{(\mathbf{v}^m)^{\mathrm{T}}\mathbf{e}_t}{\|\mathbf{v}^m\|\|\mathbf{e}_t\|}, \quad \mathbf{v}^m = \frac{1}{N}\sum_{n=1}^{N}\mathbf{v}_n^m$$

2. Generate visual-guided sentence-level embedding:

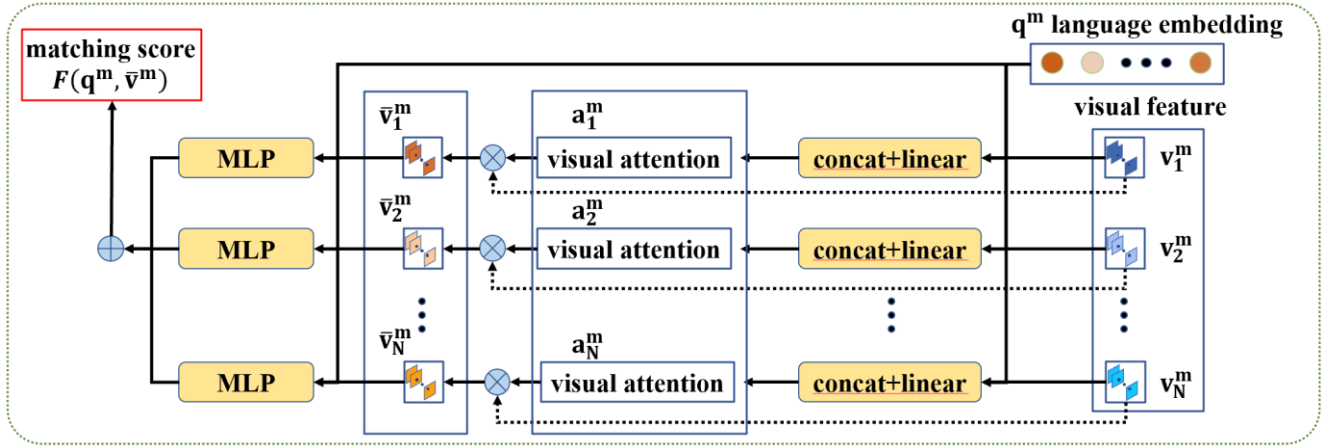
$$\bar{\mathbf{q}}^m = \sum_{t=1}^{T}\mathrm{softmax}(\lambda^m \mathbf{s}_t^m)\cdot\mathbf{e}_t$$

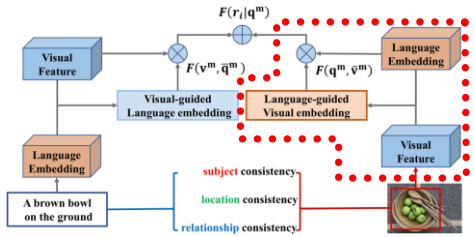
3. Calculate the score between visual feature and visual-guided language embedding:

$$F(\mathbf{v}^m, \bar{\mathbf{q}}^m) = \frac{(\mathbf{v}^m)^{\mathrm{T}}\bar{\mathbf{q}}^m}{\|\mathbf{v}^m\|\|\bar{\mathbf{q}}^m\|}$$



Language-guided visual embedding





1. Attention from language embedding to the n-th visual element of module m:

$$\mathbf{h}_n = \tanh(\mathbf{W}_1^m[\mathbf{v}_n^m, \mathbf{q}^m] + b^m),$$

$$\mathbf{a}_n^m = \mathrm{softmax}(\mathbf{W}_2^m \mathbf{h}_n)$$

2. Generate a more discriminative language-guided visual feature:

$$\bar{\mathbf{v}}^m = \sum_{n=1}^{N} \mathbf{a}_n^m \cdot \mathbf{v}_n^m$$

3. Calculate the score between language-guided visual feature and the language embedding:

$$F(\mathbf{q}^m, \bar{\mathbf{v}}^m) = \mathrm{MLP}(\mathbf{q}^m, \bar{\mathbf{v}}^m)$$

Matching result

Sum the mutual guidance scores in three modules(subject, location and relationship) as final score.

$$F(r_i|\mathbf{q}^{\mathrm{m}}) = F(\mathbf{v}^m, \overline{\mathbf{q}}^m) + F(\mathbf{q}^{\mathrm{m}}, \overline{\mathbf{v}}^{\mathrm{m}}) \ , \quad \mathrm{m} \in (\mathrm{subj}, \mathrm{loc}, \mathrm{rel})$$

$$F(r_i|\mathcal{E}) = \omega_{subj} F\left(r_i \middle| \mathbf{q}^{subj}\right) + \omega_{loc} F\left(r_i \middle| \mathbf{q}^{loc}\right) + \omega_{rel} F(r_i|\mathbf{q}^{rel})$$

Loss function

For each positive pair $(\mathcal{R}_i, \mathcal{E}_i)$, construct two negative pairs $\left(\mathcal{R}_i, \mathcal{E}_j\right)$ and $\left(\mathcal{R}_j, \mathcal{E}_i\right)$, and then calculate rank loss.

$$\mathrm{Loss} = \sum_i \left(\left[k - F(\mathcal{R}_i, \mathcal{E}_i) + F\left(\mathcal{R}_i, \mathcal{E}_j\right)\right]_+ + \left[k - F(\mathcal{R}_i, \mathcal{E}_i) + F\left(\mathcal{R}_j, \mathcal{E}_i\right)\right]_+ \right), \ [x]_+ = \max(x, 0)$$

Experiment result

Comparison with state-of-the-art methods on RefCOCO, RefCOCO+ and RefCOCOg.

Method	Box	RefCOCO			RefCOCO+			RefCOCOg		
		val	testA	testB	val	testA	testB	val*	val	test
visdif+MMI [6]	gt	-	73.98	76.59	-	59.17	55.62	64.02	-	-
Speaker/visdif [6]	gt	76.18	74.39	77.30	58.64	61.29	56.24	59.40	-	-
S-L-R [3]	gt	79.56	78.65	80.22	62.26	64.60	59.62	72.63	71.65	71.92
VC [9]	gt	-	78.98	82.39	-	62.56	62.90	73.98	-	-
Attr [21]	gt	-	78.05	78.07	-	61.47	57.22	69.83	-	-
Accu-Att [22]	gt	81.27	81.17	80.01	65.56	68.76	60.63	73.18	-	-
PLAN [23]	gt	81.67	80.81	81.32	64.18	66.31	61.46	69.47	-	-
Multi-hop Film [24]	gt	84.9	87.4	83.1	73.8	**78.7**	65.8	71.5	-	-
MattNet [8]	gt	85.65	85.26	84.57	71.01	75.13	66.17	-	78.10	78.12
NMT_{REE} [25]	gt	85.65	85.63	85.08	72.84	75.74	67.62	78.03	78.57	78.21
LGRANS [19]	gt	82.0	81.2	84.0	66.6	67.6	65.5	-	75.4	74.7
DGA [20]	gt	86.34	86.64	84.79	73.56	78.31	**68.15**	-	80.21	**80.26**
MutAtt	gt	**86.58**	**87.20**	**85.38**	**73.69**	76.30	67.74	-	**80.37**	79.24
S-L-R [3]	det	69.48	73.71	64.96	55.71	60.74	48.80	-	60.21	59.63
PLAN [23]	det	-	75.31	65.52	-	61.34	50.86	58.03	-	-
MattNet [8]	det	76.40	80.43	69.28	64.93	70.26	56.00	-	66.67	67.01
LGRANS [19]	det	-	76.6	66.4	-	64.0	53.4	62.5	-	-
DGA [20]	det	-	78.42	65.53	-	69.07	51.99	-	-	63.28
MutAtt	det	**78.35**	**82.52**	**71.50**	**67.90**	**72.60**	**58.60**	-	**68.67**	**69.03**

Ablation study

Ablation study of our method on RefCOCOg dataset.

		val	test
1	MutAtt:subj+loc+rel	77.96	77.14
2	MutAtt:subj(V→L)+loc+rel	79.33	78.53
3	MutAtt:subj(V→L+L→V)+loc+rel	80.00	79.34
4	MutAtt:subj(V⇌L)+loc(V⇌L)+rel	80.35	79.03
5	MutAtt:subj(V⇌L)+loc(V⇌L)+rel(V⇌L)	80.37	79.24

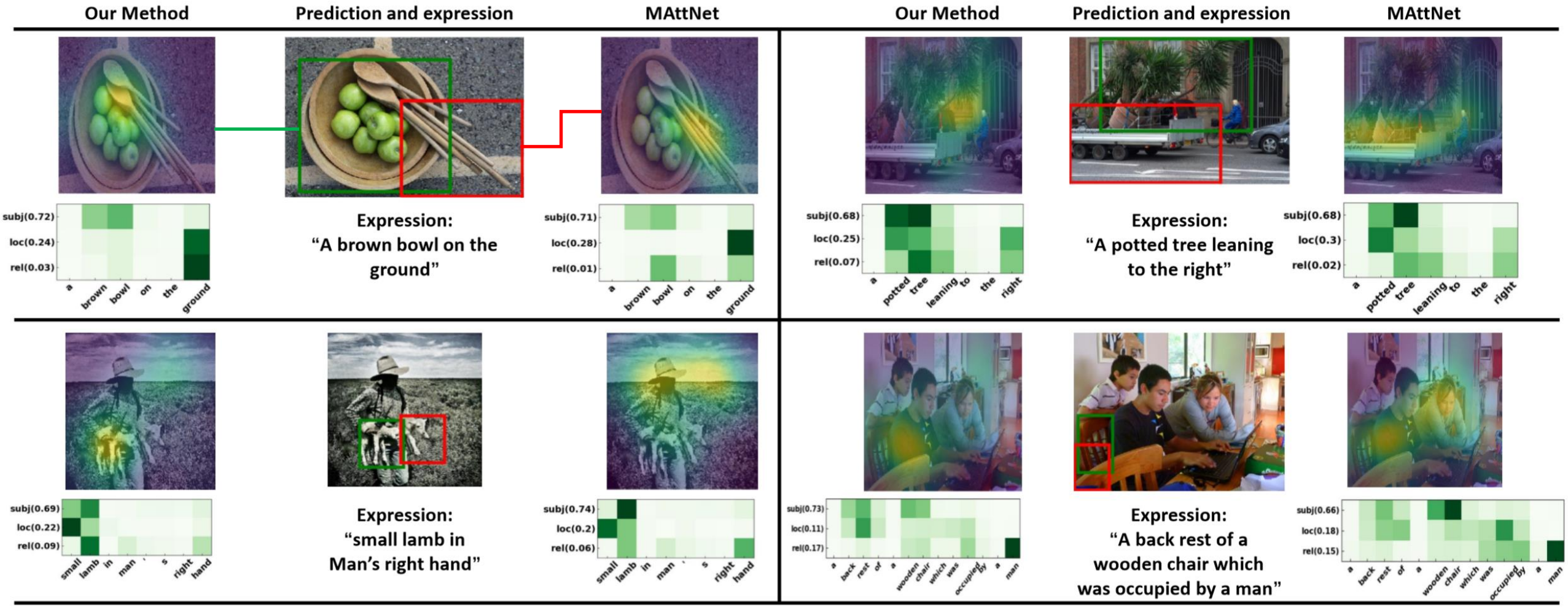
$V \rightarrow L$: add visual guidance on module m.

$L \rightarrow V$: add language guidance on module m.

$V \rightleftharpoons L$: add visual guidance and language guidance on module m.

Visualization result

Visualization comparisons between MutAtt and MAttNet of visual attention and language attention of each word on three modules.



Conclusion

- We proposed a mutual attention-based guidance method (MutAtt) for the task of REC.
- MutAtt can learn more discriminative feature while ensuring that visual information and language information play the same important role.
- Experiments on three datasets with two settings show that MutAtt outperforms other methods.

Thanks!

Q&A